In [2]:
```
#install numpy module
```

In [3]:
```
pip install numpy
```

```
Requirement already satisfied: numpy in c:\users\kvr\anaconda3\lib\site-packa
ges (1.21.5)
Note: you may need to restart the kernel to use updated packages.
```

In [4]:
```
pip list
```

```
poyo                          0.5.0
prometheus-client             0.14.1
prompt-toolkit                3.0.20
Protego                       0.1.16
psutil                        5.9.0
ptyprocess                    0.7.0
py                            1.11.0
pyasn1                        0.4.8
pyasn1-modules                0.2.8
pycodestyle                   2.8.0
pycosat                       0.6.3
pycparser                     2.21
pyct                          0.4.8
pycurl                        7.45.1
PyDispatcher                  2.0.5
pydocstyle                    6.1.1
pyerfa                        2.0.0
pyflakes                      2.4.0
Pygments                      2.11.2
PyHamcrest                    2.0.2
```

In [5]:
```
pip show numpy
```

```
Name: numpy
Version: 1.21.5
Summary: NumPy is the fundamental package for array computing with Python.
Home-page: https://www.numpy.org (https://www.numpy.org)
Author: Travis E. Oliphant et al.
Author-email:
License: BSD
Location: c:\users\kvr\anaconda3\lib\site-packages
Requires:
Required-by: astropy, bkcharts, bokeh, Bottleneck, daal4py, datashader, datas
hape, gensim, h5py, holoviews, hvplot, imagecodecs, imageio, matplotlib, mkl-
fft, mkl-random, numba, numexpr, pandas, patsy, pyerfa, PyWavelets, scikit-im
age, scikit-learn, scipy, seaborn, statsmodels, tables, tifffile, xarray
Note: you may need to restart the kernel to use updated packages.
```

In [8]:
```
#Take the array of elements in Python
import sys
lst1=[10,20,30,40]
print(lst1,type(lst1))
print("Total Memory of list object=",sys.getsizeof(lst1))
```

```
[10, 20, 30, 40] <class 'list'>
Total Memory of list object= 120
```

In [9]:
```
import numpy as np
print(np.__version__)
```

```
1.21.5
```

In [11]:
```
a=np.array(lst1)
print(a,type(a))
print("Total Memory of ndarray object=",sys.getsizeof(a))
```

```
[10 20 30 40] <class 'numpy.ndarray'>
Total Memory of ndarray object= 120
```

In [12]:
```
lst1=[10,20]
print(lst1,type(lst1))
print("Total Memory of list object=",sys.getsizeof(lst1))
```

```
[10, 20] <class 'list'>
Total Memory of list object= 72
```

In [13]:
```
a=np.array(lst1)
print(a,type(a))
print("Total Memory of ndarray object=",sys.getsizeof(a))
```

```
[10 20] <class 'numpy.ndarray'>
Total Memory of ndarray object= 112
```

In [14]:
```
lst1=[10,20,30,40,50,60,70,80,90]
print(lst1,type(lst1))
print("Total Memory of list object=",sys.getsizeof(lst1))
```

```
[10, 20, 30, 40, 50, 60, 70, 80, 90] <class 'list'>
Total Memory of list object= 152
```

In [15]:
```
a=np.array(lst1)
print(a,type(a))
print("Total Memory of ndarray object=",sys.getsizeof(a))
```

```
[10 20 30 40 50 60 70 80 90] <class 'numpy.ndarray'>
Total Memory of ndarray object= 140
```

In [16]:
```
lst1=[10,20,30,40,50,60,70,80,90,100,200,300,40,50,600,70,800,900]
print(lst1,type(lst1))
print("Total Memory of list object=",sys.getsizeof(lst1))
```

```
[10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 200, 300, 40, 50, 600, 70, 800, 90
0] <class 'list'>
Total Memory of list object= 216
```

In [17]:
```
a=np.array(lst1)
print(a,type(a))
print("Total Memory of ndarray object=",sys.getsizeof(a))
```

```
[ 10  20  30  40  50  60  70  80  90 100 200 300  40  50 600  70 800 900] <cl
ass 'numpy.ndarray'>
Total Memory of ndarray object= 176
```

In [18]:
```
a
```

Out[18]:
```
array([ 10,  20,  30,  40,  50,  60,  70,  80,  90, 100, 200, 300,  40,
        50, 600,  70, 800, 900])
```

In [20]:
```
lst1=[10,20,30]
print(lst1,type(lst1))
```

```
[10, 20, 30] <class 'list'>
```

In [21]:
```
lst1=lst1+1 # Vector Based Operations on list not Possible
```

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_668\2218900554.py in <module>
----> 1 lst1=lst1+1 # Vector Based Operations

TypeError: can only concatenate list (not "int") to list
```

In [26]:
```
#convert lst1 into ndarray object
a=np.array(lst1)
print(a,type(a))
```

```
[10 20 30] <class 'numpy.ndarray'>
```

In [27]:
```
a=a+1 # Vector Based Operations are Possible on ndarray object
a
```

Out[27]:
```
array([11, 21, 31])
```

In [29]:
```
#Data Retrieval from list object
lst=[10,20,30,40,50,60,70,80,90]
print(lst,type(lst),id(lst))
```

```
[10, 20, 30, 40, 50, 60, 70, 80, 90] <class 'list'> 1373011613696
```

In [30]:
```
for val in lst:
    print(val,id(val),id(lst))
```

```
10 1372906416720 1373011613696
20 1372906417040 1373011613696
30 1372906417360 1373011613696
40 1372906417680 1373011613696
50 1372906418000 1373011613696
60 1372906606800 1373011613696
70 1372906607120 1373011613696
80 1372906607440 1373011613696
90 1372906607760 1373011613696
```

In [32]:
```
#Data Retrieval from ndarray object
lst=[10,20,30,40,50,60,70,80,90]
a=np.array(lst)
print(a,type(a),id(a))
```

```
[10 20 30 40 50 60 70 80 90] <class 'numpy.ndarray'> 1373011525520
```

In [33]:
```
for val in a:
    print(val,id(val),id(a))
```

```
10 1373011380720 1373011525520
20 1373011689616 1373011525520
30 1373011380720 1373011525520
40 1373011689616 1373011525520
50 1373011380720 1373011525520
60 1373011689616 1373011525520
70 1373011380720 1373011525520
80 1373011689616 1373011525520
90 1373011380720 1373011525520
```

In [34]:
```
lst=[10,20,30,40,50,60,70,80,90]
print(lst,type(lst))
```

```
[10, 20, 30, 40, 50, 60, 70, 80, 90] <class 'list'>
```

In [35]:
```
lst.reshape(3,3)
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_668\707250098.py in <module>
----> 1 lst.reshape(3,3)

AttributeError: 'list' object has no attribute 'reshape'
```

In [36]:
```
lst=[10,20,30,40,50,60,70,80,90]
print(lst,type(lst))
a=np.array(lst)
print(a,type(a))
```

```
[10, 20, 30, 40, 50, 60, 70, 80, 90] <class 'list'>
[10 20 30 40 50 60 70 80 90] <class 'numpy.ndarray'>
```

In [37]:
```
a.reshape(3,3)
```

Out[37]:
```
array([[10, 20, 30],
       [40, 50, 60],
       [70, 80, 90]])
```

In [38]:
```
lst=[10,20,30,40,50,60,70,80,90,15,25,35]
print(lst,type(lst))
a=np.array(lst)
print(a,type(a))
```

```
[10, 20, 30, 40, 50, 60, 70, 80, 90, 15, 25, 35] <class 'list'>
[10 20 30 40 50 60 70 80 90 15 25 35] <class 'numpy.ndarray'>
```

In [39]:
```
a.reshape(4,3)
```

Out[39]:
```
array([[10, 20, 30],
       [40, 50, 60],
       [70, 80, 90],
       [15, 25, 35]])
```

In [40]:
```
a.reshape(3,4)
```

Out[40]:
```
array([[10, 20, 30, 40],
       [50, 60, 70, 80],
       [90, 15, 25, 35]])
```

In [41]:
```
a.reshape(6,2)
```

Out[41]:
```
array([[10, 20],
       [30, 40],
       [50, 60],
       [70, 80],
       [90, 15],
       [25, 35]])
```

In [42]:
```
a.reshape(2,6)
```

Out[42]:
```
array([[10, 20, 30, 40, 50, 60],
       [70, 80, 90, 15, 25, 35]])
```

In [43]:
```
a.reshape(2,3,2)
```

Out[43]:
```
array([[[10, 20],
        [30, 40],
        [50, 60]],

       [[70, 80],
        [90, 15],
        [25, 35]]])
```

In [44]:
```
a.reshape(2,2,3)
```

Out[44]:
```
array([[[10, 20, 30],
        [40, 50, 60]],

       [[70, 80, 90],
        [15, 25, 35]]])
```

In [45]:
```
a.reshape(4,4) # cannot reshape array of size 12 into shape (4,4)
```

```
---------------------------------------------------------------------------
ValueError                                Traceback (most recent call last)
~\AppData\Local\Temp\ipykernel_668\4045685779.py in <module>
----> 1 a.reshape(4,4)

ValueError: cannot reshape array of size 12 into shape (4,4)
```

In [46]:
```
print(a,type(a))
```

```
[10 20 30 40 50 60 70 80 90 15 25 35] <class 'numpy.ndarray'>
```

In [47]:
```
a.ndim
```

Out[47]:
```
1
```

In [48]:
```
a.shape
```

Out[48]:
```
(12,)
```

In [49]:
```
a.shape=(3,4)
```

In [50]:
```
a
```

Out[50]:
```
array([[10, 20, 30, 40],
       [50, 60, 70, 80],
       [90, 15, 25, 35]])
```

In [51]:
```
a.ndim
```
Out[51]:
```
2
```

In [52]:
```
a.shape
```
Out[52]:
```
(3, 4)
```

In [53]:
```
a.shape=(2,3,2)
```

In [54]:
```
a
```
Out[54]:
```
array([[[10, 20],
        [30, 40],
        [50, 60]],

       [[70, 80],
        [90, 15],
        [25, 35]]])
```

In [55]:
```
a.ndim
```
Out[55]:
```
3
```

In [56]:
```
a.shape
```
Out[56]:
```
(2, 3, 2)
```

In [57]:
```
lst=[10,"Rossum",23.45,True]
print(lst,type(lst))
```
```
[10, 'Rossum', 23.45, True] <class 'list'>
```

In [58]:
```
a=np.array(lst)
print(a,type(a))
```
```
['10' 'Rossum' '23.45' 'True'] <class 'numpy.ndarray'>
```

In [59]:
```
print(lst,type(lst))
```
```
[10, 'Rossum', 23.45, True] <class 'list'>
```

In [60]:
```
a=np.array(lst,object)
print(a,type(a))
```
```
[10 'Rossum' 23.45 True] <class 'numpy.ndarray'>
```

In [61]:
```
a.dtype
```
Out[61]:
```
dtype('O')
```

In [62]:
```
print(a.dtype)
```
```
object
```

In [63]:
```
a.shape=(2,2)
```

In [64]:
```
a
```
Out[64]:
```
array([[10, 'Rossum'],
       [23.45, True]], dtype=object)
```

In [65]:
```
lst=[10,"Rossum",23.45,True]
print(lst,type(lst))
```
```
[10, 'Rossum', 23.45, True] <class 'list'>
```

In [66]:
```
lst[0]=100
```

In [67]:
```
print(lst,type(lst))
```
```
[100, 'Rossum', 23.45, True] <class 'list'>
```

In [68]:
```
a=np.array(lst)
print(a,type(a))
```
```
['100' 'Rossum' '23.45' 'True'] <class 'numpy.ndarray'>
```

In [69]:
```
a[2]=99.99
```

In [70]:
```
print(a,type(a))
```
```
['100' 'Rossum' '99.99' 'True'] <class 'numpy.ndarray'>
```

In [71]:
```
l1=[[10,20,30],[40,50,60],[70,80,90]]
l2=[[1,2,3],[4,5,6],[7,8,9]]
print(l1,type(l1))
print(l2,type(l2))
```
```
[[10, 20, 30], [40, 50, 60], [70, 80, 90]] <class 'list'>
[[1, 2, 3], [4, 5, 6], [7, 8, 9]] <class 'list'>
```

In [72]:
```
a=np.array(l1)
b=np.array(l2)
print(a,type(a))
print(b,type(b))
```

```
[[10 20 30]
 [40 50 60]
 [70 80 90]] <class 'numpy.ndarray'>
[[1 2 3]
 [4 5 6]
 [7 8 9]] <class 'numpy.ndarray'>
```

In [73]:
```
c=a+b
print(c,type(c))
```

```
[[11 22 33]
 [44 55 66]
 [77 88 99]] <class 'numpy.ndarray'>
```

In [ ]: